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Vehicle height control system

Abstract

A vehicle height control system includes an arm including a first end portion connected to a vehicle body and a second end portion coupled to a wheel, a bearing unit fixed to the vehicle body, a crank coupled to the bearing unit, a push rod including a first end portion connected to the crank and a second end portion connected to the arm, and a spring reaction force variable device coupled to the bearing unit and configured to vary reaction force applied to the crank.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4540197	12/1984	Finn	280/124.167	F16F 1/26
6671596	12/2002	Kawashima	280/5.515	B60G 7/006
10703199	12/2019	Gruschow	N/A	B60G 3/04
2005/0275181	12/2004	MacIsaac	180/271	B62D 39/00
2015/0111489	12/2014	Chopra	N/A	N/A
2018/0224020	12/2017	Ohashi	N/A	N/A
2021/0293293	12/2020	Krause	N/A	N/A
2024/0100898	12/2023	Chang	N/A	B60G 3/18

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3050623	12/2017	CA	B60G 15/068
204506399	12/2014	CN	N/A
111417530	12/2022	CN	B60G 11/107
102010033002	12/2011	DE	B60G 11/183
102013002713	12/2013	DE	B60G 11/15
102013012637	12/2014	DE	B60G 11/50
102016213195	12/2017	DE	N/A
1698495	12/2005	EP	B60G 11/181
2228240	12/2009	EP	B60G 11/16
1894756	12/2017	EP	B60G 17/0162
2468487	12/2009	GB	B60G 11/16
H 08-178119	12/1995	JP	N/A
2846413	12/1998	JP	N/A
2009132237	12/2008	JP	B60G 17/0157
2016-023684	12/2015	JP	N/A
2024165846	12/2023	JP	N/A
2000-0055777	12/1999	KR	N/A
20020052281	12/2001	KR	N/A
2006-0112245	12/2005	KR	N/A
20070060480	12/2006	KR	N/A

10-1339218	12/2013	KR	N/A
10-2379149	12/2021	KR	N/A
2022-0096698	12/2021	KR	N/A
WO-2014124721	12/2013	WO	B60G 11/181
WO-2015010773	12/2014	WO	B60G 11/50
WO-2017079263	12/2016	WO	B60G 3/04
WO-2019177486	12/2018	WO	B60G 17/08
WO-2021240415	12/2020	WO	B60G 13/14

OTHER PUBLICATIONS

C. Lee and S. Oh, “Configuration and performance analysis of a compact planetary geared Elastic Actuator,” IECON 2016—42nd Annual Conference of the IEEE Industrial Electronics Society, 2016, pp. 6391-6396. cited by applicant

Lee, C.; Kwak, S.; Kwak, J.; Oh, S. Generalization of Series Elastic Actuator Configurations and Dynamic Behavior Comparison. Actuators 2017, 6, 26. cited by applicant

Arnaldo Gomes Leal Junior, et al., “Series Elastic Actuator: Design, Analysis and Comparison” Chapter 10, 2016. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korean Patent Application No. 10-2023-0028900, filed on Mar. 6, 2023, the entire contents of which is incorporated herein for all purposes by this reference.

BACKGROUND OF THE PRESENT DISCLOSURE

Field of the Present Disclosure

(2) The present disclosure relates to a vehicle height control system. More particularly, it relates to a vehicle height control system capable of controlling vehicle height and roll behavior of a vehicle by controlling input torque of a separate spring reaction force variable device depending on the load state of the vehicle.

Description of Related Art

(3) Generally, a vehicle and a means of transportation use a suspension including a spring device having elastic force and a damper device having absorbing force, and the suspension serves as a device configured to maintain vehicle height, which is the vertical distance of the highest point of the vehicle from the ground plane, to support the weight of the vehicle, and to mitigate the impact the vehicle receives from the ground.

(4) When a vehicle having a high center of gravity, such as a large bus or a double-decker bus, is traveling on the road, overturning of the vehicle may occur due to a rapid rolling motion caused by sudden steering. To reduce the risk of such an accident, it is required not only to change damping force of the suspension but also to adjust vehicle height by lowering a vehicle body.

(5) On the other hand, when a vehicle having low vehicle height is traveling on the road, the lower side of a vehicle body or a bumper may contact with a speed bump or a protruding portion on the road surface. Therefore, a vehicle height adjustment device configured to adjust the vehicle height is provided in a vehicle having low vehicle height. The vehicle height adjustment device controls

the vehicle height by driving a motor. However, generally, because power is continuously supplied to the motor to prevent back drive of the motor after vehicle height adjustment is performed by the vehicle height adjustment device, energy consumption of the vehicle may increase. Accordingly, it is necessary to solve a problem related to energy consumption.
(6) Meanwhile, the vehicle height adjustment device, generally, adjusts, depending on the load state of a vehicle occupant or baggage, the vehicle height within the range of load conditions of curb weight and gross vehicle weight (GVW). That is, a change in adjustable payload is limited. Recently, as types of vehicles become more diverse, a purpose built vehicle (refer to hereinafter as PBV) has emerged on the market. In the case of a detachable transporter among PBVs, when a business box (a main body) is detached from the transporter in the forward-and-rearward direction, there is a significant difference between a payload to be supported when the transporter exists independently and a payload to be supported when the same has the business box attached thereto. Therefore, there is a need for a technique capable of controlling the vehicle height depending on a much wider range of vehicle load states.
(7) Meanwhile, while a vehicle is traveling on the road, vibrations such as rolling, pitching, and yawing occur in addition to vertical bouncing of the vehicle body. In the instant case, a suspension system is used to reliably absorb these vibrations to improve ride comfort as well as steering stability. Among the vibrations, the movement of the vehicle body from side to side is referred to as rolling, and when the present rolling occurs, ride comfort and driving stability of the vehicle may deteriorate and deformation of the vehicle body may occur. For the present reason, the suspension system of the vehicle is provided with a device configured to prevent rolling of the vehicle body.
(8) For example, the suspension system includes a stabilizer bar configured to prevent a vehicle from shaking from side to side and to maintain parallelism of a vehicle body. Here, the stabilizer bar controls roll behavior of a vehicle by torsion generated by a phase difference between a bump stroke and a rebound stroke of a left or right wheel of the vehicle. However, such a vehicle roll control device has a limitation in controlling roll behavior generated by active movement of a vehicle due to a characteristic of the stabilizer bar configured to control roll behavior of a vehicle. Accordingly, there is a problem in that it is impossible to provide both stability and ride comfort of a vehicle only using a stabilizer bar including a fixed rigidity.
(9) The information included in this Background in this Background of the present disclosure is only for enhancement of understanding of the general background of the present disclosure and may not be taken as an acknowledgement or any form of suggestion that this information forms the prior art already known to a person skilled in the art.

BRIEF SUMMARY

(10) Various aspects of the present disclosure are directed to providing a vehicle height control system configured for controlling vehicle height and roll behavior of a vehicle by controlling input torque of a separate spring reaction force variable device depending on the load state of the vehicle.
(11) Furthermore, various aspects of the present disclosure are directed to providing a vehicle height control system configured for maintaining vehicle height in a state in which continuous driving force of a motor is not applied through a clutch unit located in a spring reaction force variable device.
(12) The objects of the present disclosure are not limited to the above-mentioned objects, and other technical objects not mentioned herein will be clearly understood by those skilled in the art from the detailed description of the embodiments. Furthermore, the objects of the present disclosure may be achieved by means indicated in the scope of the claims and a combination thereof.
(13) Various aspects of the present disclosure are directed to providing a vehicle height control system including an arm including a first end portion connected to a vehicle body and a second end portion coupled to a wheel, a bearing unit fixed to the vehicle body, a crank coupled to the bearing unit, a push rod including a first end portion connected to the crank and a second end portion connected to the arm, and a spring reaction force variable device coupled to the bearing unit and

configured to vary reaction force applied to the crank.

(14) In an exemplary embodiment of the present disclosure, the spring reaction force variable device may further include a motor configured to provide torque, a clutch unit connected to an output shaft of the motor, a reducer located at an output end portion of the clutch unit and coupled to the crank, and a spring portion coupled to the reducer. Reaction torque of the spring portion may be varied by driving the motor.

(15) In another exemplary embodiment of the present disclosure, in the spring reaction force variable device, when the motor is driven, motor torque may be applied to the reducer coupled to the output end portion so that the motor torque on the reducer, the reaction torque of the spring portion, and load torque by an external load applied to the reducer through the crank are balance, and when the motor is not driven, the reaction torque and the load torque may be in balance.

(16) In various exemplary embodiments of the present disclosure, the clutch unit may rotate only when the torque is input through the output shaft of the motor.

(17) In various exemplary embodiments of the present disclosure, the reducer may be formed of a planetary gear set, and the planetary gear set may include a sun gear coupled to the output end portion of the clutch unit, a ring gear fixed to the spring portion, a plurality of planetary gears located and engaged between the sun gear and the ring gear, and a carrier including a first end portion connected to the planetary gears and a second end portion connected to the bearing unit.

(18) In still various exemplary embodiments of the present disclosure, the spring portion may further include a frame fixed to the vehicle body, an elastic portion including one end portion fixed to the frame, and a ring gear fixing portion including the other end portion of the elastic portion fixed thereto, the ring gear fixing portion being coupled to the ring gear.

(19) In a further exemplary embodiment of the present disclosure, the elastic portion may include a first elastic portion located adjacent to the frame and a second elastic portion located adjacent to the ring gear.

(20) In another further exemplary embodiment of the present disclosure, the bearing unit may include a bearing case fixed to the vehicle body, a rod portion formed to be integrated with the crank and supported by the bearing case, a spring coupling portion located at one end portion of the rod portion and coupled to a torsion spring, and a variable device coupling portion located at the other end portion of the rod portion and coupled to the spring reaction force variable device.

(21) In yet another further exemplary embodiment of the present disclosure, the bearing case may further include a support bearing configured to rotatably support the rod portion.

(22) In yet another further exemplary embodiment of the present disclosure, the torsion spring and the spring coupling portion may be coupled to each other through a tooth-meshing structure or a bolt fastening structure.

(23) In still yet another further exemplary embodiment of the present disclosure, an output end portion of the spring reaction force variable device and the variable device coupling portion may be coupled to each other through a tooth-meshing structure or a bolt fastening structure.

(24) Various aspects of the present disclosure are directed to providing a vehicle height control system including an arm including a first end portion connected to a vehicle body and a second end portion coupled to a wheel, a bearing unit fixed to the vehicle body, a crank coupled to the bearing unit, a push rod including a first end portion connected to the crank and a second end portion connected to the arm, a spring reaction force variable device coupled to the bearing unit and configured to vary reaction force applied to the crank, and a control unit configured to perform, when a measured vehicle height exceeds a preset vehicle height range, driving of the spring reaction force variable device.

(25) In an exemplary embodiment of the present disclosure, the spring reaction force variable device may further include a motor configured to provide torque, a clutch unit connected to an output shaft of the motor, a reducer located at an output end portion of the clutch unit and coupled to the crank, and a spring portion coupled to the reducer. Reaction torque of the spring portion may

be varied by driving the motor.
(26) In another exemplary embodiment of the present disclosure, the control unit may perform driving of the motor when the measured vehicle height exceeds the preset vehicle height range.
(27) In various exemplary embodiments of the present disclosure, the control unit may be configured to determine an amount of change in the vehicle height according to each driving of the motor, and to set the number of times of the driving of the motor based on the determined amount of change in the vehicle height.
(28) In various exemplary embodiments of the present disclosure, the reducer may be formed of a planetary gear set, and the planetary gear set may include a sun gear coupled to the output end portion of the clutch unit, a ring gear fixed to the spring portion, a plurality of planetary gears located to move between the sun gear and the ring gear, and a carrier including a first end portion connected to the planetary gears and a second end portion connected to the bearing unit.
(29) In still various exemplary embodiments of the present disclosure, the control unit may convert a target control amount of the vehicle height into a target rotation amount of the carrier and may drive the motor so that the carrier is rotated up to a target location.
(30) In a further exemplary embodiment of the present disclosure, the bearing unit may include a bearing case fixed to the vehicle body, a rod portion formed to be integrated with the crank and supported by the bearing case, a spring coupling portion located at one end portion of the rod portion and coupled to a torsion spring, and a variable device coupling portion located at the other end portion of the rod portion and coupled to the spring reaction force variable device.
(31) Other aspects and exemplary embodiments of the present disclosure are discussed infra.
(32) It is understood that the terms “vehicle”, “vehicular”, and other similar terms as used herein are inclusive of motor vehicles in general, such as passenger vehicles including sports utility vehicles (SUV), buses, trucks, various commercial vehicles, watercraft including a variety of boats and ships, aircraft, and the like, and include hybrid vehicles, electric vehicles, plug-in hybrid electric vehicles, hydrogen-powered vehicles, and other alternative fuel vehicles (e.g., fuels derived from resources other than petroleum). As referred to herein, a hybrid vehicle is a vehicle that has two or more sources of power, for example, vehicles powered by both gasoline and electricity.
(33) The above and other features of the present disclosure are discussed infra.
(34) The methods and apparatuses of the present disclosure have other features and advantages which will be apparent from or are set forth in more detail in the accompanying drawings, which are incorporated herein, and the following Detailed Description, which together serve to explain certain principles of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view exemplarily illustrating a vehicle height control system according to various exemplary embodiments of the present disclosure;
(2) FIG. **2** is a diagram illustrating a bearing unit of the vehicle height control system according to the exemplary embodiment of the present disclosure:
(3) FIG. **3** is a diagram illustrating a detailed configuration of a spring reaction force variable device of an independent corner module according to the exemplary embodiment of the present disclosure:
(4) FIG. **4**A is a diagram illustrating a configuration of a spring portion of the vehicle height control system according to the exemplary embodiment of the present disclosure:
(5) FIG. **4**B is a diagram illustrating another configuration of the spring portion of the vehicle height control system according to the exemplary embodiment of the present disclosure:
(6) FIG. **4**C is a diagram illustrating a driving relationship in which torque balance is achieved

when a motor is driven according to the exemplary embodiment of the present disclosure:
(7) FIG. **4**D is a driving relationship in which torque balance is achieved when the motor is not driven according to the exemplary embodiment of the present disclosure;
(8) FIG. **5**A is a diagram illustrating a configuration of a clutch unit of the vehicle height control system according to the exemplary embodiment of the present disclosure:
(9) FIG. **5**B is an enlarged view of a braking portion and a steel portion of the clutch unit according to the exemplary embodiment of the present disclosure;
(10) FIG. **5**C is a diagram illustrating driving force applied to a locker in a state in which driving force of a motor output shaft is applied according to the exemplary embodiment of the present disclosure;
(11) FIG. **5**D illustrates a block diagram of the clutch unit in a state in which the driving force of the motor output shaft is applied according to the exemplary embodiment of the present disclosure;
(12) FIG. **5**E illustrates an operation diagram in which an output end portion thereof is rotated in a state in which the driving force of the motor output shaft is applied according to the exemplary embodiment of the present disclosure; and
(13) FIG. **6** illustrates a schematic diagram of the clutch unit in a state in which driving force of the output end portion thereof is applied according to the exemplary embodiment of the present disclosure.
(14) It should be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various exemplary features illustrative of the basic principles of the present disclosure. The specific design features of the present disclosure as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes will be determined in part by the particular intended application and use environment.
(15) In the figures, reference numbers refer to the same or equivalent portions of the present disclosure throughout the several figures of the drawing.
DETAILED DESCRIPTION
(16) Hereinafter, reference will be made in detail to various embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings and described below. While the present disclosure will be described in conjunction with exemplary embodiments of the present disclosure, it will be understood that present description is not intended to limit the present disclosure to the exemplary embodiments of the present disclosure. On the other hand, the present disclosure is directed to cover not only the exemplary embodiments of the present disclosure, but also various alternatives, modifications, equivalents, and other embodiments, which may be included within the spirit and scope of the present disclosure as defined by the appended claims. The exemplary embodiments are provided to more completely describe the present disclosure to those skilled in the art.
(17) Furthermore, terms such as “part” and “unit” described in the specification refer to a unit configured to process at least one function or operation, and the unit may be implemented by hardware or a combination of pieces of hardware.
(18) Furthermore, in the present specification, when it is said that any part is positioned “on” or “above” another part, it means that the part is “directly on” the other part. In the instant case, another part may be positioned between the two parts. Furthermore, when it is said that any part is positioned “under” or “below” another part, it means that the part is “directly under” the other part. In the instant case as well, another part may be positioned between the two parts.
(19) Furthermore, a control unit **1600** of the present specification may be implemented by an algorithm configured to control the operation of various components disposed in a vehicle, a memory configured to store data about a program that reproduces the algorithm, and a processor configured to perform the above-described operation using data stored in the memory. In the instant case, the memory and the processor may be implemented as separate chips. Alternatively, the memory and the processor may be implemented as a single chip. For example, the control unit

1600 may include at least one of an electronic control unit (ECU), a central processing unit (CPU), a microprocessor unit (MPU), a microcontroller unit (MCU), an application processor (AP), or any type of processor well known in the technical field of the present disclosure. Furthermore, the control unit **1600** may include at least one application configured to execute a method according to the exemplary embodiments of the present disclosure, or the same may be formed of a combination of software and hardware configured for performing an arithmetic operation on a program.

(20) Furthermore, in each step, an identification code is used for convenience of description, and the identification code does not describe the order of each step. Each step may be performed in a different order from the order described in the exemplary embodiments unless a specific order is explicitly stated in the context.

(21) Additionally, in the following embodiments, a reducer may be described with substantially the same configuration as a planetary gear set, and may be interpreted as including the planetary gear set as a type of reducer.

(22) Furthermore, a wheel **2000** of a vehicle is described in the exemplary embodiment of the present disclosure, and each wheel **2000** of a multi-wheeled vehicle may move independently.

(23) Furthermore, "height" included in the exemplary embodiment of the present disclosure may mean a distance between the center portion of the wheel **2000** and a vehicle body.

(24) Various embodiments of the present disclosure relate to a vehicle height control system, and more particularly to a system configured for controlling the height of a vehicle body by controlling torque applied to a crank **1200** in response to torque of a motor **1510** located in a spring reaction force variable device **1500**.

(25) FIG. **1** is a diagram illustrating a configuration of a vehicle height control system according to various exemplary embodiments of the present disclosure.

(26) As illustrated in the drawing, the vehicle height control system includes an arm **1000** including one end portion fixed to a vehicle body and the other end portion coupled to the wheel **2000**. Here, the arm **1000** is coupled to the wheel **2000** through a knuckle, and the other end portion of the arm **1000** coupled to the wheel **2000** is configured to move in the vertical direction with respect to the one end portion coupled to the vehicle body.

(27) Furthermore, the arm **1000** includes a bearing unit **1100** fixed to the vehicle body and the crank **1200** located in a direction perpendicular to the longitudinal direction of a rod portion **1120** formed in the bearing unit **1100**. The bearing unit **1100** includes a torsion spring **1400** and the spring reaction force variable device **1500** respectively coupled to the opposite end portions of the rod portion **1120**. A spring coupling portion **1130** and a variable device coupling portion **1140** are respectively disposed at the opposite end portions of the rod portion **1120** of the bearing unit **1100**, and the torsion spring **1400** and the spring reaction force variable device **1500** are respectively coupled to the spring coupling portion **1130** and the variable device coupling portion **1140**. The rod portion **1120** is located through a bearing case **1110** of the bearing unit **1100**. The crank **1200** is formed to be integrated with the rod portion **1120** and is located in the bearing case **1110**. Furthermore, the bearing case **1110** may further include a support bearing **1121** configured to be rotatable through the rod portion **1120**. The support bearing **1121** is fixed to the vehicle body so that the rod portion **1120** rotates freely.

(28) The vehicle height control system includes a push rod **1300** located between the end portion of the crank **1200** and the end portion of the arm **1000**. The push rod **1300** is configured to move integrally with the end portion of the crank **1200** in response to the vertical movement of the arm **1000**. Furthermore, torque applied to the crank **1200** is transmitted to the rod portion **1120** according to the movement direction of the arm **1000**. The opposite end portions of the push rod **1300** are respectively coupled to the crank **1200** and the arm **1000** through a bearing, and the push rod **1300** is configured to move integrally with the arm **1000** in the vertical direction thereof. That is, when a change in vehicle height is applied, a torque is applied to the torsion spring **1400** and the spring reaction force variable device **1500** through the crank **1200**, and torque is applied to the

crank **1200** in response to driving of the spring reaction force variable device **1500**, adjusting the vehicle height.

(29) When the currently measured vehicle height via sensors exceeds a vehicle height range preset in the control unit **1600**, the control unit **1600** is configured to drive the motor **1510** to apply torque to the crank **1200** so that the vehicle height is adjusted, and to control a driving amount of the motor **1510** so that the vehicle height is adjusted within the preset vehicle height range.

(30) The control unit **1600** may set a driving frequency of the motor **1510** so that the current vehicle height is controlled to be adjusted within the vehicle height range set in the control unit **1600**. That is, driving of the motor **1510** may be performed a plurality of times in response to a change in vehicle height which is variable in response to each driving of the motor **1510**.

(31) A torque balance relationship of the spring reaction force variable device **1500** according to driving of the motor **1510** is included in FIG. **4**C below, and a torque balance relationship of the spring reaction force variable device **1500** after driving of the motor **1510** stops is included in FIG. **4**D. According to the torque balance relationships included in FIGS. **4**C and **4**D, each torque balance relationship is newly set depending on the driving frequency of the motor **1510**.

(32) The spring reaction force variable device **1500** coupled to the other side of the bearing unit **1100** is configured to rotate a carrier **1525** of the spring reaction force variable device **1500** by driving force of the motor **1510**. The driving force of the motor **1510** is converted into reaction torque of the carrier **1525** of the spring reaction force variable device **1500**, and the reaction torque is transmitted to the crank **1200** directly coupled to the rod portion **1120** of the bearing unit **1100**. The variable device coupling end portion of the bearing unit **1100** and the carrier **1525** may be bolted together, or the variable device coupling end portion and the carrier **1525** may be coupled to each other by a tooth-meshing structure.

(33) The spring reaction force variable device **1500** is coupled to the rod portion **1120** of the bearing unit **1100**, and the torsion spring **1400** may apply torque to the rod portion **1120** according to driving of the motor **1510**. Furthermore, the spring reaction force variable device **1500** is configured to provide torque to an end portion of the crank **1200** formed to be integrated with the rod portion **1120**, the end portion being coupled to the push rod **1300**. Furthermore, when torque is applied to the spring reaction force variable device **1500** through the rod portion **1120**, the spring reaction force variable device **1500** is configured to rotate the carrier **1525** to achieve torque balance between the applied torque, reaction force of the spring reaction force variable device **1500**, and reaction torque of the torsion spring **1400**. Accordingly, the crank **1200** may be rotated integrally with the push rod **1300** in the height direction of the vehicle in response to the driving amount and driving frequency of the motor **1510** of the spring reaction force variable device **1500**.

(34) FIG. **2** is an enlarged view of the bearing unit **1100** according to the exemplary embodiment of the present disclosure. As illustrated in the drawing, the bearing unit **1100** includes the bearing case **1110** coupled to a vehicle body, and the bearing case **1110** is located so that the rod portion **1120** is provided to penetrate the bearing case **1110** in the longitudinal direction of the vehicle. Furthermore, the bearing case **1110** through which the rod portion **1120** passes may include a support bearing configured to support the rod portion **1120**.

(35) Furthermore, the vehicle height control system includes the crank **1200** fixed to the rod portion **1120** and located in the width direction of the vehicle. Accordingly, the other far end portion of the crank **1200** may be moved in the height direction of the vehicle in response to the rotation amount of the rod portion **1120**. The crank **1200** is coupled to the central area between the support bearings located on the rod portion **1120**, and the other end portion of the crank **1200** is moved integrally with the rod portion **1120** in the height direction in response to the rotation amount of the rod portion **1120**. The arm **1000** and the wheel **2000** are moved integrally in response to the movement of the crank **1200** in the height direction thereof.

(36) The opposite end portions of the rod portion **1120** are respectively coupled to the torsion spring **1400** and the carrier **1525** of the spring reaction force varying device **1500**. The torsion

spring **1400** may be located in parallel to the rod portion **1120** in the same direction, and the torsion spring **1400** may be fixed to the rod portion **1120** through the spring coupling portion **1130** located at one end portion of the rod portion **1120**.
(37) The other end portion of the rod portion **1120** includes the variable device coupling portion **1140** configured to allow the carrier **1525** of the spring reaction force variable device **1500** to be coupled to the other end portion of the rod portion **1120**. That is, the driving torque of the motor **1510**, the reaction torque of the torsion spring **1400**, and the reaction torque applied from the spring reaction force variable device **1500** are transmitted to the crank **1200** through the rod portion **1120**. Furthermore, the torque applied through the crank **1200** and the reaction torque of the torsion spring **1400** are applied to the spring reaction force variable device **1500** through the rod portion **1120**.
(38) The torsion spring **1400** and the carrier **1525** respectively coupled to the opposite end portions of the rod portion **1120** may be fixed to the rod portion **1120** through bolt fastening. Alternatively, a meshing gear may be provided at each of the opposite end portions of the spring coupling portion **1130** and the variable device coupling portion **1140**, and each of the torsion spring **1400** and the carrier **1525** may include a structure corresponding to the meshing gear formed in each of the spring coupling portion **1130** and the variable device coupling portion **1140** to be meshed with each other.
(39) That is, the rod portion **1120** is coupled to the torsion spring **1400** and the spring reaction force variable device **1500** so that torque applied from each of the torsion spring **1400** and the spring reaction force variable device **1500** is transmitted to the crank **1200**.
(40) FIG. **3** is a diagram illustrating the configuration of the spring reaction force variable device **1500** according to the exemplary embodiment of the present disclosure.
(41) The spring reaction force variable device **1500** is fixed to the vehicle body, and the carrier **1525** of the spring reaction force variable device **1500** is coupled to one end portion of the rod portion **1120** of the bearing unit **1100**. Furthermore, the spring reaction force variable device **1500** includes the motor **1510** configured to apply torque, a clutch unit **10** coupled to an output shaft of the motor **1510**, and a reducer **1520** coupled to the output end portion **300** of the clutch unit **10**.
(42) The reducer **1520** is rotated in response to torque applied from the output end portion **300** of the clutch unit **10** coupled to the reducer **1520**. Furthermore, a spring portion **1530** is provided between the vehicle body and the outside of the reducer **1520**. That is, the spring portion **1530** has one end portion coupled to a ring gear **1523** located at the edge portion of the reducer **1520** and the other end portion fixed to the vehicle body. Furthermore, the spring portion **1530** is configured to apply reaction torque applied between the vehicle body and the ring gear **1523** to the carrier **1525** of the reducer **1520**.
(43) That is, torque applied from the motor **1510** is transmitted to the reducer **1520** through the clutch unit **10**, and the reducer **1520** is configured to determine the location of the carrier **1525** of the reducer **1520** based on resultant force of elastic force applied from the spring portion **1530**. Therefore, a positional relationship between components forming the reducer **1520** is set to achieve torque balance between the reducer **1520**, the carrier **1525**, the motor **1510**, and the rod portion **1120** coupled to the carrier **1525**.
(44) According to the exemplary embodiment of the present disclosure, the reducer **1520** is formed of a planetary gear set, and the same includes a sun gear **1521** coupled to the output end portion **300** of the clutch unit **10** and located at the center portion thereof, and a ring gear **1523** configured to surround the outermost portion of the planetary gear set. Furthermore, a plurality of planetary gears **1522** located between the ring gear **1523** and the sun gear **1521** are provided. The external surface of the planetary gear set includes the carrier **1525**, coupled to the planetary gear **1522** and rotated integrally with the planetary gear **1522**, and the carrier **1525** is coupled to the rod portion **1120** of the bearing unit **1100**. The carrier **1525** is inserted into the center portion of the planetary gear **1522** and is rotated integrally with the planetary gear **1522**, and the carrier **1525** may be

located substantially on the same axis as the output end portion **300** of the clutch unit **10**.
(45) That is, while torque is applied to the sun gear **1521** from the output end portion **300** of the clutch unit **10** and the planetary gear **1522** is rotated in the opposite direction to the sun gear **1521** in response to rotation of the sun gear **1521**, the carrier **1525** is rotated in the same direction as the sun gear **1521** between the ring gear **1523** and the sun gear **1521**. Furthermore, when the planetary gear **1522** is rotated integrally with the carrier **1525** as described above, the ring gear **1523** is configured to rotate at a predetermined angle with respect to the vehicle body, and an elastic portion **1531** of the spring portion **1530** is configured to provide reaction torque to the ring gear **1523** rotating with respect to a frame **1540**.
(46) The spring portion **1530** includes the frame **1540** fixed to the vehicle body, and the same is coupled to a ring gear fixing portion **1524** located on the ring gear **1523** of the reducer **1520**. When the ring gear **1523** is rotated, the elastic portion **1531** located between the frame **1540** and the ring gear fixing portion **1524** is configured to provide reaction torque between the frame **1540** and the ring gear **1523** in response to the rotation amount of the ring gear **1523**. Accordingly, the reaction torque applied through the spring portion **1530** in response to the rotation amount of the ring gear **1523** may be relatively increased.
(47) When the motor **1510** is rotated to drive the sun gear **1521**, the elastic portion **1531** is configured to apply reaction torque to the ring gear **1523** in response to the rotation amount of the ring gear **1523**. Conversely, when the motor **1510** is not rotated, the elastic portion **1531** may provide reaction torque in response to the rotation amount between the ring gear **1523** and the frame **1540** by external torque applied to the carrier **1525**. That is, the elastic portion **1531** provides reaction torque in the opposite direction to the external torque introduced into the carrier **1525** to achieve torque balance of external torque introduced into the spring reaction force variable device **1500**.
(48) The elastic portion **1531** may include one or more elastic portions **1531** having different spring constants corresponding to the deformation amount and reaction torque, and as various exemplary embodiments of the present disclosure, the elastic portion **1531** may include a torsional spring, a coil spring, and the torsion spring **1400**. Moreover, the elastic portion **1531** may be formed of a combination of one or more different spring types.
(49) FIG. **4**A illustrates, as various exemplary embodiments of the present disclosure, a configuration in which a first elastic portion **1532** and a second elastic portion **1533** forming the elastic portion **1531** are coupled to each other in series to apply reaction torque to the ring gear **1523**.
(50) As illustrated in the drawing, the frame **1540** is fixedly located so as not to be rotated, and includes the first elastic portion **1532** located adjacent to the frame **1540**. The first elastic portion **1532** is coupled to the frame **1540** through a spring connection portion **1534** located in the frame **1540**. In the exemplary embodiment of the present disclosure, the spring connection portion **1534** may be coupled to the edge portion of the first elastic portion **1532**.
(51) Furthermore, the second elastic portion **1533** located adjacent to the ring gear **1523** is provided. The second elastic portion **1533** is coupled to the ring gear fixing portion **1524** through the spring connection portion **1534**. The spring connection portion **1534** may be located adjacent to the edge portion of the second elastic portion **1533** coupled to the ring gear **1523**.
(52) Furthermore, the first elastic portion **1532** and the second elastic portion **1533** may include the spring connection portion **1534** to be coupled to each other at a location adjacent to the central area. That is, the first elastic portion **1532** and the second elastic portion **1533** are connected in series.
(53) Therefore, when the motor **1510** is driven, the ring gear **1523** is rotated by rotation force applied to the sun gear **1521** coupled to the output end portion **300** of the clutch unit **10**, and the frame **1540** and the ring gear fixing portion **1524** are switched to a relatively twisted state, and reaction torque of the elastic portion **1531** is generated.
(54) Furthermore, when the motor **1510** is not driven, the output end portion **300** of the clutch unit

10 is fixed to limit the rotation of the sun gear **1521**, and reaction torque of the elastic portion **1531** coupled to the ring gear fixing portion **1524** and external torque of a load applied to the carrier **1525** are applied to the reducer **1520**. In the instant case, the elastic portion **1531** provides reaction torque in a direction opposite to the twisting direction. Accordingly, the carrier **1525** is rotated by an angle relatively smaller than the amount of torque applied from the outside thereof to achieve torque balance.
(55) FIG. **4**B illustrates, as another exemplary embodiment of the present disclosure, a configuration in which the first elastic portion **1532** and the second elastic portion **1533** forming the elastic portion **1531** are coupled in parallel to each other to apply reaction torque to the ring gear **1523**.
(56) As illustrated in the drawing, the elastic portion **1531** includes the first elastic portion **1532** located adjacent to the frame **1540** and the second elastic portion **1533** located adjacent to the side surface of the ring gear **1523**. The first elastic portion **1532** and the second elastic portion **1533** are coupled to each other by the spring connection portion **1534** formed to extend from the side of the ring gear **1523**. The spring connection portion **1534** located on the side of the ring gear **1523** is coupled to the edge portion areas of the first elastic portion **1532** and the second elastic portion **1533**.
(57) Furthermore, the spring connecting portion **1534** formed to extend from the frame **1540** may be fixedly located at the central areas of the first elastic portion **1532** and the second elastic portion **1533**.
(58) In another exemplary embodiment of the present disclosure, the elastic portion **1531** is configured to apply reaction torque between the ring gear **1523** and the frame **1540** when the motor **1510** is driven and the motor **1510** is not driven, making it possible to achieve, according to the driving relationship, torque balance between torque applied to the carrier **1525**, torque generated by driving the sun gear **1521**, and reaction torque applied to the ring gear **1523** through the elastic portion **1531**.
(59) FIG. **4**C is a diagram illustrating a mechanism configured to achieve torque balance in response to a driving input of the motor **1510**.
(60) The control unit **1600** is configured to apply power to the motor **1510**, and when the output shaft of the motor **1510** is rotated, the clutch unit **10** coupled to the output shaft of the motor **1510** is rotated integrally with the motor **1510**. The reducer **1520** coupled to the output end portion **300** of the clutch unit **10** is configured to rotate the carrier **1525** by driving force of the output end portion **300** of the clutch unit **10**.
(61) The sun gear **1521** coupled to the output end portion **300** of the clutch unit **10** is rotated in the same direction as the output end portion **300** of the clutch unit **10**, and each of the planetary gears **1522** located to be engaged with the sun gear **1521** is rotated in the opposite direction to the sun gear **1521**. Furthermore, each of the planetary gears **1522** moves between the sun gear **1521** and the ring gear **1523**, and the central axis of the planetary gear **1522** is configured to move in the same direction as the rotation direction of the sun gear **1521**. Furthermore, the carrier **1525**, formed to protrude from the external surface to be connected to the planetary gear **1522** and coupled to a load, is rotated in the same direction as the movement direction of the central axis of the planetary gear **1522** and the rotation direction of the sun gear **1521**.
(62) Additionally, when the ring gear **1523** is rotated relative to the frame **1540** in response to the movement of the planetary gear **1522**, torsional force is applied to the spring portion **1530** located between the frame **1540** and the ring gear **1523**. Accordingly, the elastic portion **1531** of the spring portion **1530** is configured to apply reaction torque to the ring gear **1523** in response to relative torsion between the frame **1540** and the ring gear fixing portion **1524**.
(63) Here, FIG. **4**C illustrates a balance relationship after torque balance is achieved between the outputs of the carrier **1525** coupled to the outside in response to the input of the motor **1510**, the reducer **1520**, and the motor **1510**.

(64) As illustrated in the drawing, torque of the sun gear **1521** acts counterclockwise when the motor **1510** is driven, and reaction torque of the spring portion **1530** coupled to the ring gear **1523** acts in the same direction as the torque of the sun gear **1521**. Conversely, reaction torque or torque applied from the carrier **1525** acts clockwise. Therefore, in torque balance, the sum of the torque of the motor **1510** and the reaction torque of the spring portion **1530** is in balance with the torque of the sun gear **1521**.
(65) That is, when force applied from the motor **1510** to the sun gear **1521** is determined as Ts and reaction torque of the spring portion **1530** is applied in a state in which the ring gear **1523** in torque balance is fixed, torque Tc applied to the carrier **1525** is determined by multiplying 1+R (where R is the radius of the ring gear/the radius of the sun gear) by the force Ts applied to the sun gear **1521**. Furthermore, reaction torque Tr of the ring gear **1523** has R*Ts.
(66) As described above, as illustrated in the drawing, when torque balance is achieved in response to the input of the motor **1510**, Tc=Tr+Ts is obtained, and when reaction torque is applied in a state in which the ring gear **1523** is fixed, a torque balance state including a value of Tc=(1+R)*Ts is achieved.
(67) As various exemplary embodiments of the present disclosure, FIG. **4**D illustrates movement to achieve an balance state, when driving force of the motor **1510** is released, between the reaction torque Tc of the carrier **1525** through the spring reaction force variable device **1500** and the reaction torque Tr applied from the outside of the spring reaction force variable device **1500** in a state in which rotation of the output end portion **300** of the clutch unit **10** is limited.
(68) Here, the driving force of the motor **1510** is released, and torque applied to the sun gear **1521** is 0, which means a torque balance state between the reaction torque of the spring portion **1530** and the torque applied to the carrier **1525**.
(69) As illustrated in the drawing, in the state in which the driving of the motor **1510** of the spring reaction force variable device **1500** is released, the output end portion **300** coupled to the sun gear **1521** is switched to a state in which driving torque is not applied from the motor **1510**. Accordingly, torque Tc of the carrier **1525** applied counterclockwise in the drawing is generated, and reaction torque Tr applied through the spring portion **1530** of the spring reaction force variable device **1500** is applied. Here, reaction torque applied from the outside of the spring reaction force variable device **1500** may include all external factors affecting the spring reaction force variable device **1500** through the carrier **1525**.
(70) That is, when the input torque of the motor **1510** is converted to 0 after torque Ts of the motor **1510** is applied, the sun gear **1521** is fixed by the clutch unit **10**, and the carrier **1525** connected to the planetary gear **1522** is rotated to obtain an equilibrium state between Tc and Tr. For example, in the relationship of Tc>Tr, a relationship is established in which the carrier **1525** is rotated counterclockwise in the drawing and restored by the reaction torque Tc, and Tr applied by the spring portion **1530** increases. Here, in the torque balance relationship of the total planetary gears **1522**, new torque Tc_new of the carrier **1525** and new torque Tr_new of the spring portion **1530** are determined and represented as Tc_new=Tr_new in the balanced state.
(71) That is, Ts torque to be applied when the motor **1510** is driven in FIG. **4**C becomes 0, and the Ts torque is configured to achieve a new balanced state through torque applied to the spring portion **1530** and torque output through the carrier **1525**. Accordingly, torque Ts/2 is applied to each of the spring portion **1530** and the carrier **1525**. In response to the torque Ts/2 applied thereto, a positional relationship of the ring gear **1523** coupled to the carrier **1525** and the spring portion **1530** is adjusted.
(72) Wherein the $Tc=\theta c\times K1$ and $Tr=\theta r\times K2$
(73) Additionally, according to a relationship between elastic modulus K2 of the spring portion **1530** and elastic modulus K1 applied to the carrier **1525**, the carrier **1525** and the planetary gear **1522** are moved to a new balanced position to be rotated clockwise in the drawing.
(74) That is, the control unit **1600** may be configured to determine the rotation amount (angle) of

the carrier **1525** and the planetary gear **1522** that achieve a new torque balance after driving of the motor **1510** is stopped, and to set the driving frequency of the motor **1510** based on the determined rotation amount. Furthermore, the control unit **1600** may set the driving frequency of the motor **1510** based on the rotation amount of the carrier **1525** rotated in response to each driving of the motor **1510**, making it possible to determine the rotation amount of the carrier **1525**.
(75) According to the exemplary embodiment of the present disclosure, the control unit **1600** applies driving force of the motor **1510** as illustrated in FIG. **4**C, is configured to determine the rotation amount of the carrier **1525** by performing the step in FIG. **4**D to obtain new torque balance based on the applied driving force, and repeatedly performs, based on the determined rotation amount, the steps in FIGS. **4**C and **4**D to correspond to the predetermined rotation amount of the carrier **1525**.
(76) FIG. **5**A is a diagram illustrating a configuration relationship of the clutch unit **10** according to the exemplary embodiment of the present disclosure.
(77) As illustrated in the drawing, the clutch unit **10** includes a housing **100** and a cover portion **110** disposed at one end portion of the housing **100** and configured to cover the open end portion of the housing **100**. The housing **100** includes a circular cross section, and the cover portion **110** is configured to completely cover the opening of one end portion of the housing **100**.
(78) The clutch unit **10** further includes an output end portion **300** configured to penetrate the other end portion of the housing **100** and formed to have at least one flat surface. The housing **100** includes a plurality of lockers **400** disposed therein and configured to surround the flat surface of the output end portion **300** and a motor output shaft **200** disposed therein, the motor output haft **200** including one end portion inserted into an opening **410** located in each locker **400**. The output end portion **300** includes the number of flat surfaces corresponding to the number of lockers **400** located inside the housing **100**. The output end portion **300** in an exemplary embodiment of the present disclosure may have four flat surfaces corresponding to the four lockers **400**. Furthermore, the plurality of flat surfaces located at the output end portion **300** are respectively in contact with the plurality of adjacent lockers **400** when torque of the motor output shaft **200** is applied. Accordingly, the surfaces of the lockers **400** and the flat surfaces of the output end portion **300** may selectively contact with each other.
(79) The motor output shaft **200** includes a rotation transmission portion **210** disposed at one end portion thereof and partially inserted into the opening **410** located in each locker **400** in the longitudinal direction and a driving transmission portion **220** disposed at the other end portion thereof and formed to penetrate the cover portion **110** to protrude outwards from the cover portion **110**. The driving transmission portion **220** is coupled to a driving portion configured to apply torque to be rotated integrally with the rotation direction of the driving portion.
(80) Furthermore, torque of the driving portion is applied to the driving transmission portion **220** located at the other end portion of the motor output shaft **200**, and driving force applied to the driving transmission portion **220** is configured to rotate the output end portion **300** through the rotation transmission portion **210**. The driving portion is configured to transmit driving force capable of rotating the motor output shaft **200**, and the surfaces of the plurality of lockers **400** are respectively in contact with the flat surfaces of the output end portion **300** in response to the torque of the motor output shaft **200**. According to the exemplary embodiment of the present disclosure, the driving portion coupled to the motor output shaft **200** may be formed of the motor **1510**.
(81) The motor output shaft **200** includes the rotation transmission portion **210** inserted into the opening **410** formed in each of the lockers **400**. Here, the motor output shaft **200** includes four rotation transmission portions **210** corresponding to the four lockers **400** in the exemplary embodiment of the present disclosure. Each of the rotation transmission portions **210** may maintain a state of being inserted into each of the openings **410** formed in the plurality of lockers **400**. Additionally, the rotation transmission portion **210** is rotated in the same direction as the rotation direction of the driving transmission portion **220**, and the locker **400** in contact with the rotation

transmission portion **210** through the opening **410** is rotated integrally with the rotation transmission portion **210** in response to the rotation direction of the motor output shaft **200**.
(82) A braking unit **600** located inside the housing **100** of the clutch unit **10** is configured to regulate movement of the plurality of lockers **400** when torque of the output end portion **300** is applied to the inside of the clutch unit **10**, and the torque of the output end portion **300** is not transmitted to the motor output shaft **200**. The braking unit **600** in an exemplary embodiment of the present disclosure includes a magnetic portion **440** located on the outermost side of the locker **400**, a steel portion **500** located on the internal circumferential surface of the housing **100** and disposed at a location corresponding to the magnetic portion **440**, and a braking portion **510** disposed adjacent to the steel portion **500** and configured to selectively contact with the locker **400**.
(83) The plurality of lockers **400** are located inside the housing **100**, and each flat surface of the output end portion **300** and each locker **400** may be located adjacent to each other. The plurality of lockers **400** are divided into at least two pieces, and each locker **400** is disposed to include a predetermined gap between the flat surface of the output end portion **300** and the housing **100**. The number of flat surfaces of the output end portion **300** may be the same as the number of lockers **400** so that the internal surface of each locker **400** is located adjacent to the flat surface of the output end portion **300**.
(84) Furthermore, when the torque of the motor output shaft **200** is applied, the internal end portion of the locker **400** may contact with the flat surface formed on the output end portion **300**, and the locker **400** may be spaced from the internal circumferential surface of the housing **100** with a predetermined distance so that the motor output shaft **200**, the locker **400**, and the output end portion **300** are integrally rotated without interference with the internal circumferential surface of the housing **100**.
(85) The housing **100** includes the steel portion **500** disposed on the internal circumferential surface thereof, and at least one locker **400** includes the magnetic portion **440** disposed on the external circumferential surface thereof. Accordingly, when the torque of the motor output shaft **200** is released, the magnetic portion **440** of the locker **400** may be moved to a location close to the internal circumferential surface of the housing **100**. Furthermore, the braking unit **600** includes the braking portion **510** disposed adjacent to the steel portion **500** and located close to the internal circumferential surface of the housing **100**. Accordingly, the external circumferential surface of the locker **400** is moved to a location in contact with the braking portion **510** by magnetic force to limit the movement of the motor output shaft **200**.
(86) Furthermore, when the torque of the output end portion **300** is applied, the flat surfaces formed on the output end portion **300** push the plurality of lockers **400** in the radial direction so that the braking portion **510** located on the internal circumferential surface of the housing **100** and the external circumferential surfaces of the lockers **400** contact with each other to be fixed to each other. Therefore, it is possible to prevent the torque of the output end portion **300** from being transmitted to the motor output shaft **200**. The braking portion **510** may be formed at a location closer to the locker **400** than to the steel portion **500**, making it possible to prevent the magnetic portion **440** of the locker **400** from directly contacting with the steel portion **500**.
(87) The locker **400** and the internal circumferential surface of the housing **100** may be configured to form a predetermined gap therebetween according to the location of the locker **400**. Therefore, in a state in which driving force of the motor output shaft **200** is released, the external circumferential surface of the locker **400** is moved to a location adjacent to the internal circumferential surface of the housing **100** by magnetic force of the magnetic portion **440**, and a distance between the internal circumferential surface of the housing **100** and the external circumferential surface of the locker **400** becomes minimized.
(88) Conversely, when the motor output shaft **200** is rotated, the rotation transmission portion **210** of the motor output shaft **200** may be located to contact one end portion in the width direction of the opening **410** of the locker **400**, and torque may be applied to rotate each locker **400** in the

torque direction of the driving portion. In the instant case, the plurality of lockers **400** are located to respectively contact with the flat surfaces of the output end portion **300**, and the distance between the internal circumferential surface of the housing **100** and the external circumferential surface of the locker **400** is switched to the maximum state. Accordingly, the locker **400** is located in response to the rotation of the motor output shaft **200** to tightly contact with the surface of the output end portion **300** (that is, the surface of the output end portion **300** is constrained) without generating reaction force with the housing **100**.

(89) As described above, the clutch unit **10** in an exemplary embodiment of the present disclosure is configured so that, in response to the motor output shaft **200** configured to rotate in a direction consistent with the rotation direction applied from the driving portion, the locker **400** is spaced from the internal circumferential surface of the housing **100** and is rotated integrally with the output end portion **300**. Furthermore, when the torque applied to the motor output shaft **200** is released, the braking portion **510** and the locker **400** contact with each other to limit the movement of the motor output shaft **200**, preventing back drive of the motor.

(90) FIG. **5**B is a diagram illustrating a configuration of the steel portion **500** located on the internal circumferential surface of the housing **100** and the braking portion **510** located adjacent to the steel portion **500**.

(91) The steel portion **500** is located on the internal circumferential surface of the housing **100** to correspond to the magnetic portions **440** respectively located on the external circumferential surfaces of the plurality of lockers **400**. Therefore, when the torque of the motor output shaft **200** is released, the locker **400** is moved to a location adjacent to the steel portion **500** of the housing **100** by magnetic force. At the same time, the external surface of the locker **400** may contact with the braking portion **510** to limit the movement of the locker **400** and the motor output shaft **200**.

(92) Furthermore, the braking portion **510** located adjacent to the steel portion **500** may be located closer to the center portion of the housing **100** than the steel portion **500**. Accordingly, when the locker **400** is moved to a location adjacent to the steel portion **500** in response to magnetic force, the external surface of the locker **400** is configured to contact with the braking portion **510**.

(93) As illustrated in the drawing, the exemplary embodiment of the present disclosure may include the braking portion **510** having a predetermined step with the steel portion **500**, and the braking portion **510** may be located at at least a portion of the opposite end portions of the housing **100** in the longitudinal direction with respect to the steel portion **500**. The exemplary embodiment of the present disclosure may include the braking portion **510** located on at least a portion of the steel portion **500** in the longitudinal direction and configured to surround the internal circumferential surface of the housing **100**.

(94) Therefore, when the locker **400** is moved to a location closest to the internal circumferential surface of the housing **100**, the magnetic portion **440** and the steel portion **500** are configured to maintain a non-contact state and to provide reaction force in a state in which the braking portion **510** and the locker **400** contact with each other.

(95) FIGS. **5**C to **5**E are diagrams illustrating a configuration in which torque of the motor output shaft **200** is applied and the motor output shaft **200** is rotated integrally with the locker **400** and the output end portion **300**.

(96) As illustrated in FIG. **5**C, the motor **1510** is configured as a driving portion and is coupled to the driving transmission portion **220** of the motor output shaft **200**, and when torque of the motor **1510** is applied to the motor output shaft **200**, the rotation transmission portions **210** of the motor output shaft **200** respectively located at the openings **410** of the locker **400** are configured to initially press the openings **410** in the rotation direction of the motor output shaft **200**.

(97) The pressurized openings **410** respectively move the plurality of lockers **400** so that the external surfaces of the lockers **400** are spaced from the braking portion **510** of the housing **100**, and accordingly, the plurality of lockers **400** are spaced from the internal circumferential surface of the housing **100** and switched to the rotatable state.

(98) The opening **410** includes a trapezoidal shape, the long side surface of which is formed close to the external circumferential surface of the housing **100**. Furthermore, the inclined side surface of the trapezoidal shape is pressed by rotation force of the rotation transmission portion **210**. Force is applied to the locker **400** in a tangential direction in which the inclined side surface and the rotation transmission portion **210** contact with each other. The force applied to the inclined side surface is formed of a resultant force including a vertical force by which the locker **400** constrains the flat surface of the output end portion **300** and a horizontal force by which the locker **400** is rotated.
(99) Furthermore, as illustrated in the drawing, in the exemplary embodiment of the present disclosure including four lockers **400**, the rotation transmission portions **210** respectively inserted into the openings **410** are rotated in the same direction, and each of the rotation transmission portions **210** contacts with a corresponding one of the end portions of the openings **410** to apply rotation force to the lockers **400** in the same direction.
(100) Furthermore, the locker **400** includes a pressure protrusion **420** located at one end portion of the locker **400**, the pressure protrusion **420** being located at one end portion of the side surface of the locker **400**, the one end portion being close to the internal circumferential surface of the housing **100**. Furthermore, the locker **400** including the pressing protrusion **420** includes an insertion groove **430** configured to allow the pressure protrusion **420** to be inserted into the adjacent locker **400**. The pressure protrusion **420** is located to cross the insertion groove **430** in the longitudinal direction, and in the lockers **400** adjacent to each other, the insertion groove **430** corresponding to the pressure protrusion **420** and the pressure protrusion **420** corresponding to the insertion groove **430** are respectively formed, allowing the adjacent lockers **400** to be coupled to each other.
(101) The plurality of lockers **400** including the pressure protrusion **420** and the insertion groove **430** are configured so that the respective lockers **400** are mutually coupled to each other. Accordingly, when the locker **400** is moved adjacent to the internal circumferential surface of the housing **100** by the magnetic portion **440** located on the outermost side of at least one locker **400**, all the lockers **400** mutually coupled to each other may be integrally moved.
(102) Furthermore, when rotation force of the rotation transmission portion **210** is applied, the pressure protrusion **420** applies force to the adjacent locker **400** in a direction in which the adjacent lockers **400** respectively contact with the flat surfaces of the output end portion **300**. In the exemplary embodiment of the present disclosure, when the motor output shaft **200** is rotated, each surface of the output end portion **300** including four flat surfaces and each surface of the lockers **400** are configured to contact with each other.
(103) Therefore, when the opening **410** of the locker **400** is pressed by the rotation transmission portion **210**, the pressure protrusion **420** of the locker **400** is inserted into the insertion groove **430** of the adjacent locker **400** to be coupled to each other. That is, the pressure protrusion **420** presses the surface of the adjacent insertion groove **430**, and the locker **400**, the surface of which is pressed by the pressure protrusion **420**, is configured to move the adjacent lockers **400** in a direction of contacting with a plurality of parallel surfaces of the output end portion **300**.
(104) In the present manner, when the motor output shaft **200** is rotated, the locker **400** presses the lockers **400** adjacent to each other to apply force in the same direction as the rotation direction thereof. Furthermore, the locker **400** includes the pressure protrusion **420** and the insertion groove **430** to press the adjacent locker **400** so that the flat surface of the output end portion **300** and the internal surface of the adjacent locker **400** contact with each other.
(105) As illustrated in FIG. 5D, the locker **400** contacts with the flat surface of the output end portion **300** so that the surface of the output end portion **300** is constrained by the locker **400**, and when the surfaces of at least some of the lockers **400** adjacent to each other contact with the output end portion **300**, the flat surfaces of the output end portion **300** are located to be in contact with the plurality of lockers **400**. Furthermore, the internal circumferential surface of the housing **100** and the external circumferential surface of the locker **400** are configured to be spaced from each other.

(106) Therefore, torque applied to the rotation transmission portion **210** of the motor output shaft **200** is transmitted to the locker **400** and the output end portion **300** without interference with the internal circumferential surface of the housing **100**.
(107) As illustrated in FIG. **5**E, the motor output shaft **200**, the plurality of lockers **400**, and the output end portion **300** are integrally rotated according to the torque of the motor output shaft **200**.
(108) Accordingly, the torque of the motor output shaft **200** is transmitted to the output end portion **300** corresponding to clockwise rotation in the drawing.
(109) In the present manner, when driving of the motor **1510** is applied, as illustrated in FIG. **4**C, rotational torque of the motor output shaft **200** is applied to the sun gear **1521**, and torque balance is achieved by the applied torque of the sun gear **1521**, the reaction torque of the spring portion **1530**, and the external torque applied from the rod portion **1120** to the carrier **1525**.
(110) FIG. **6** illustrates a driving relationship of the clutch unit **10** when torque is applied to the output end portion **300** in a state in which the torque of the motor output shaft **200** is released.
(111) As illustrated in the drawing, when the output end portion **300** is rotated, force is provided to push the locker **400** located adjacent to the output end portion **300** in the radial direction of the housing **100**, and the locker **400** is configured to limit rotation thereof by contacting with the braking portion **510** located on the internal circumferential surface of the housing **100**.
(112) That is, torque by the torque applied to the output end portion **300** applies force to move the locker **400** in the radial direction of the housing **100**, and the locker **400** and the braking portion **510** are configured to contact with each other according to the applied force. Moreover, the movement of the locker **400** is limited through magnetic force formed between the magnetic portion **440** and the steel portion **500** as well as the torque of the output end portion **300**. Therefore, the torque applied to the output end portion **300** is offset by reaction force formed between the braking portion **510** and the locker **400**, and the torque introduced from the output end portion **300** is not transmitted to the motor output shaft **200**.
(113) That is, when the sun gear **1521** is rotated to apply torque to the clutch unit **10**, the locker **400** configured to surround the output end portion **300** is located to contact with the braking portion **510** so that the torque of the output end portion **300** is not transmitted to the motor output shaft **200**.
(114) As described above, when the motor **1510** is not driven, the torque applied to the clutch unit **10** through the output end portion **300** of the clutch unit **10** is not applied to the motor output shaft **200** by the braking portion **510**. Therefore, as illustrated in FIG. **4**D, the torque applied to the reducer through the motor **1510** and the torque transmitted from the reducer to the motor **1510** become 0, and new torque balance is achieved based on the reaction force of the spring portion **1530** located in the reducer and the reaction force applied to the rod portion **1120** coupled to the carrier. Furthermore, the planetary gear **1522** and the carrier **1525** are moved to a location at which the new torque balance is achieved, and the load portion **1120** is rotated in response to the rotation amount (rotation angle) of the carrier **1525**, performing control of the vehicle height.
(115) As is apparent from the above description, the present disclosure may obtain the following effects by the above-described configuration, coupling, and use relationship.
(116) Various aspects of the present disclosure are directed to providing a vehicle control system configured to control vehicle height by adjusting input torque of a separate spring reaction force variable device depending on the load state of a vehicle, making it possible to reliably control the vehicle height when a load difference is large.
(117) Furthermore, the amount of motor driving of the spring reaction force variable device and the number of times thereof are controlled to adjust suspension characteristics of a vehicle body generated depending on the driving state of the vehicle, including an effect of simultaneously improving driving stability and ride comfort of the vehicle.
(118) The aforementioned invention can also be embodied as computer readable codes on a computer readable recording medium. The computer readable recording medium is any data

storage device that can store data which may be thereafter read by a computer system and store and execute program instructions which may be thereafter read by a computer system. Examples of the computer readable recording medium include Hard Disk Drive (HDD), solid state disk (SSD), silicon disk drive (SDD), read-only memory (ROM), random-access memory (RAM), CD-ROMs, magnetic tapes, floppy discs, optical data storage devices, etc and implementation as carrier waves (e.g., transmission over the Internet). Examples of the program instruction include machine language code such as those generated by a compiler, as well as high-level language code which may be executed by a computer using an interpreter or the like.
(119) In various exemplary embodiments of the present disclosure, each operation described above may be performed by a control device, and the control device may be configured by a plurality of control devices, or an integrated single control device.
(120) In various exemplary embodiments of the present disclosure, the scope of the present disclosure includes software or machine-executable commands (e.g., an operating system, an application, firmware, a program, etc.) for enabling operations according to the methods of various embodiments to be executed on an apparatus or a computer, a non-transitory computer-readable medium including such software or commands stored thereon and executable on the apparatus or the computer.
(121) Furthermore, the terms such as “unit”, “module”, etc. included in the specification mean units for processing at least one function or operation, which may be implemented by hardware, software, or a combination thereof.
(122) In an exemplary embodiment of the present disclosure, the vehicle may be referred to as being based on a concept including various means of transportation. In some cases, the vehicle may be interpreted as being based on a concept including not only various means of land transportation, such as cars, motorcycles, trucks, and buses, that drive on roads but also various means of transportation such as airplanes, drones, ships, etc.
(123) For convenience in explanation and accurate definition in the appended claims, the terms “upper”, “lower”, “inner”, “outer”, “up”, “down”, “upwards”, “downwards”, “front”, “rear”, “back”, “inside”, “outside”, “inwardly”, “outwardly”, “interior”, “exterior”, “internal”, “external”, “forwards”, and “backwards” are used to describe features of the exemplary embodiments with reference to the positions of such features as displayed in the figures. It will be further understood that the term “connect” or its derivatives refer both to direct and indirect connection.
(124) The term “and/or” may include a combination of a plurality of related listed items or any of a plurality of related listed items. For example, “A and/or B” includes all three cases such as “A”, “B”, and “A and B”.
(125) In the present specification, unless stated otherwise, a singular expression includes a plural expression unless the context clearly indicates otherwise.
(126) In exemplary embodiments of the present disclosure, “at least one of A and B” may refer to “at least one of A or B” or “at least one of combinations of at least one of A and B”. Furthermore, “one or more of A and B” may refer to “one or more of A or B” or “one or more of combinations of one or more of A and B”.
(127) In the exemplary embodiment of the present disclosure, it should be understood that a term such as “include” or “have” is directed to designate that the features, numbers, steps, operations, elements, parts, or combinations thereof described in the specification are present, and does not preclude the possibility of addition or presence of one or more other features, numbers, steps, operations, elements, parts, or combinations thereof.
(128) According to an exemplary embodiment of the present disclosure, components may be combined with each other to be implemented as one, or some components may be omitted.
(129) The foregoing descriptions of specific exemplary embodiments of the present disclosure have been presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed, and obviously many modifications

and variations are possible in light of the above teachings. The exemplary embodiments were chosen and described in order to explain certain principles of the invention and their practical application, to enable others skilled in the art to make and utilize various exemplary embodiments of the present disclosure, as well as various alternatives and modifications thereof. It is intended that the scope of the present disclosure be defined by the Claims appended hereto and their equivalents.

Claims

1. A vehicle height control system comprising: an arm including a first end portion connected to a vehicle body and a second end portion coupled to a wheel; a bearing unit fixed to the vehicle body; a crank coupled to the bearing unit; a push rod including a first end portion connected to the crank and a second end portion connected to the arm; and a spring reaction force variable device coupled to the bearing unit and configured to vary reaction force applied to the crank, wherein the spring reaction force variable device further includes: a motor configured to provide torque; a clutch unit connected to an output shaft of the motor; a reducer located at an output end portion of the clutch unit and coupled to the crank; and a spring portion coupled to the reducer, and wherein reaction torque of the spring portion is varied by driving the motor.
2. The vehicle height control system of claim 1, wherein, in the spring reaction force variable device, in response that the motor is driven, motor torque is applied to the reducer coupled to the output end portion so that the motor torque on the reducer, the reaction torque of the spring portion, and load torque by an external load applied to the reducer through the crank are balanced, and in response that the motor is not driven, the reaction torque and the load torque are balanced.
3. The vehicle height control system of claim 1, wherein the clutch unit rotates only when the torque of the motor is input through the output shaft of the motor.
4. The vehicle height control system of claim 1, wherein the reducer is formed of a planetary gear set, and wherein the planetary gear set includes: a sun gear coupled to the output end portion of the clutch unit; a ring gear fixed to the spring portion; a plurality of planetary gears located and engaged between the sun gear and the ring gear; and a carrier including a first end portion connected to the planetary gears and a second end portion connected to the bearing unit.
5. The vehicle height control system of claim 4, wherein the spring portion further includes: a frame fixed to the vehicle body; an elastic portion including a first end portion fixed to the frame; and a ring gear fixing portion to which a second end portion of the elastic portion is fixed and to which the ring gear is coupled.
6. The vehicle height control system of claim 5, wherein the elastic portion includes: a first elastic portion located adjacent to the frame; and a second elastic portion located adjacent to the ring gear.
7. The vehicle height control system of claim 6, further including a spring connection portion connecting the first elastic portion and the second elastic portion in series or in parallel to each other.
8. The vehicle height control system of claim 5, wherein the ring gear fixing portion is coupled to an edge portion of the elastic portion, and the frame is coupled to a central area of the elastic portion.
9. The vehicle height control system of claim 1, wherein the bearing unit includes: a bearing case fixed to the vehicle body; a rod portion formed to be integrated with the crank and supported by the bearing case; a spring coupling portion located at a first end portion of the rod portion and coupled to a torsion spring; and a variable device coupling portion located at a second end portion of the rod portion and coupled to the spring reaction force variable device.
10. The vehicle height control system of claim 9, wherein the bearing case further includes a support bearing configured to rotatably support the rod portion.
11. The vehicle height control system of claim 9, wherein the torsion spring and the spring

coupling portion are coupled to each other through a tooth-meshing structure or a bolt fastening structure.

12. The vehicle height control system of claim 9, wherein an output end portion of the spring reaction force variable device and the variable device coupling portion are coupled to each other through a tooth-meshing structure or a bolt fastening structure.

13. A vehicle height control system comprising: an arm including a first end portion connected to a vehicle body and a second end portion coupled to a wheel; a bearing unit fixed to the vehicle body; a crank coupled to the bearing unit; a push rod including a first end portion connected to the crank and a second end portion connected to the arm; a spring reaction force variable device coupled to the bearing unit and configured to vary reaction force applied to the crank; and a control unit electrically connected to the spring reaction force variable device and configured to perform, in response that a measured vehicle height exceeds a preset vehicle height range, driving of the spring reaction force variable device, wherein the spring reaction force variable device further includes: a motor configured to provide torque; a clutch unit connected to an output shaft of the motor; a reducer located at an output end portion of the clutch unit and coupled to the crank; and a spring portion coupled to the reducer, and wherein reaction torque of the spring portion is varied by driving the motor.

14. The vehicle height control system of claim 13, wherein the control unit is configured to perform driving of the motor in response that the measured vehicle height exceeds the preset vehicle height range.

15. The vehicle height control system of claim 14, wherein the control unit is configured to determine an amount of change in the vehicle height according to each driving of the motor, and to set a number of times of the driving of the motor based on the determined amount of change in the vehicle height.

16. The vehicle height control system of claim 14, wherein the control unit is configured to convert a target control amount of the vehicle height into a target rotation amount of the carrier and to drive the motor so that the carrier is rotated up to a target location.

17. The vehicle height control system of claim 13, wherein the reducer is formed of a planetary gear set, and wherein the planetary gear set includes: a sun gear coupled to the output end portion of the clutch unit; a ring gear fixed to the spring portion; a plurality of planetary gears located and engaged between the sun gear and the ring gear; and a carrier including a first end portion connected to the planetary gears and a second end portion connected to the bearing unit.

18. The vehicle height control system of claim 13, wherein the bearing unit includes: a bearing case fixed to the vehicle body; a rod portion formed to be integrated with the crank and supported by the bearing case; a spring coupling portion located at a first end portion of the rod portion and coupled to a torsion spring; and a variable device coupling portion located at a second end portion of the rod portion and coupled to the spring reaction force variable device.
